Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks Available					
5 HT	(a)	(i)		All 4 points plotted correctly ± <1 small square division (1) 1-3 points plotted correctly ± <1 small square division (0) Straight line of best fit between 1.0 – 6.0 V within ± <1 small square division of each point (1)		2		2	2	2
		(ii)		As the voltage increases, the current increases (1) at a decreasing rate (1) Award 2 marks - equal increases of voltage produce smaller increases in current **Don't accept** in a non-proportional way / the increase gets less steep / non-linearly		2		2	2	2
	(b)	(i)		Substitution: $\frac{3.0}{0.85}$ (1) = 3.53 [Ω] (1) Accept 3.5 [Ω] **Don't accept** 3.52 [Ω]	1	1		2	2	2
		(ii)		0.85 (1) **i.e. denominator from (b)(i)** + 0.6 **ecf** from graph = 1.45 [A] (1)		2		2	2	2
		(iii)		Substitution and recognition of total current I: $R = \frac{V}{I} = \frac{3.0}{1.45\ \textbf{ecf}}$ (1) = 2.07 [Ω] (1) Accept 2.1 or 2 [Ω] **Alternative:** $R_{\text{wire}} = \frac{V}{I} = \frac{3.0}{0.6\ \textbf{ecf}} = 5$ (1) then $\frac{1}{3.53\ \textbf{ecf}} + \frac{1}{5\ \textbf{ecf}} = 0.483$ so R_{total} = 2.07 [Ω] (1) Accept 2.1 or 2 [Ω]		2		2	2	2
		(iv)		By having two loops from the battery there is more current (1) Greater current implies smaller resistance (1)	2			2		2

Question				Marking details	Marks Available					
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		(v)		The current through the lamp is smaller than the current through the resistor or values stated 1.2 [A] for the resistor and 1.125 [A] for the lamp (1) [Since the voltage is the same] bigger resistance results in a smaller current (1) **Alternative: calculations based** R = 5 [Ω] for wire (1) and R = 5.33 [Ω] for lamp (1) Don't accept reference to gradients		2		2	2	2
		(vi)		**Any 2 × (1) from:** • At 5.50 V or where the lines cross [the currents are equal so] the powers are equal • [$I_{resistor}$ is less than I_{lamp}] up to 5.50 V, therefore P_{lamp} > $P_{resistor}$ (since voltages are the same and $P=VI$) • Beyond 5.50 V, [$I_{resistor}$ is greater than I_{lamp} therefore] P_{lamp} < $P_{resistor}$		2		2	2	2
				Question 5 total	**3**	**13**	**0**	**16**	**14**	**16**